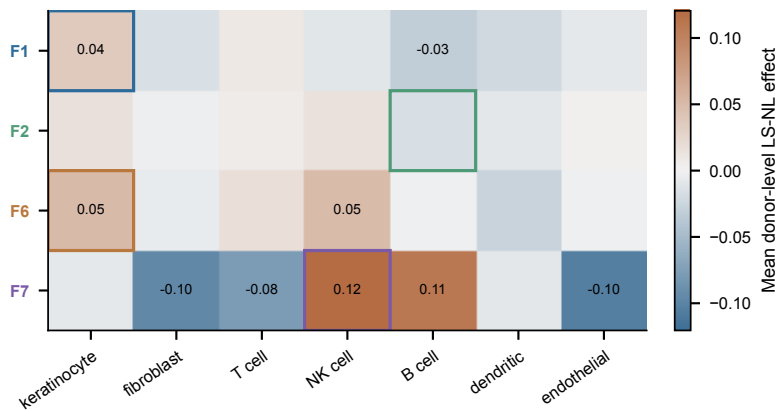


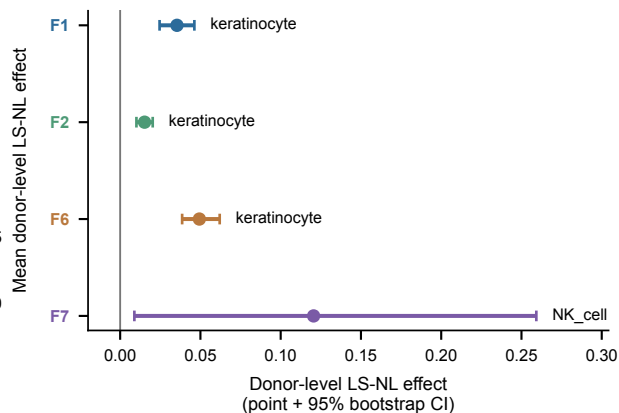
Directional cellular and spatial contextualization of retained molecular programs

a Cell-type-specific lesional-non-lesional program shifts

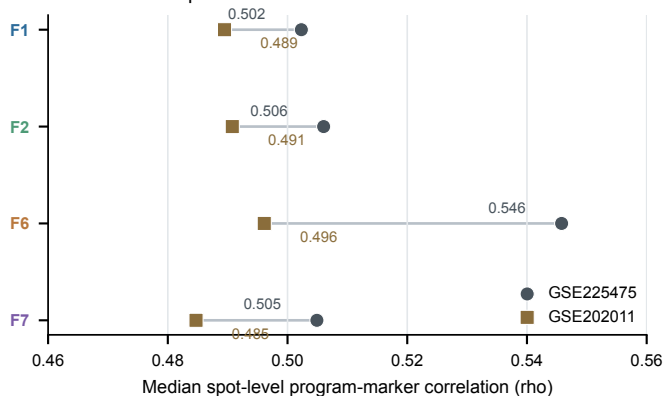


Labels shown for |effect| ≥ 0.03 ; outline marks largest absolute shift per program.

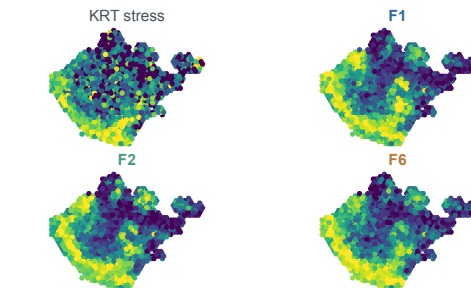
b Donor-level directional effects



c Cross-dataset spatial concordance



d Representative spatial distribution



PP1; real Visium spot coordinates; separate display scales